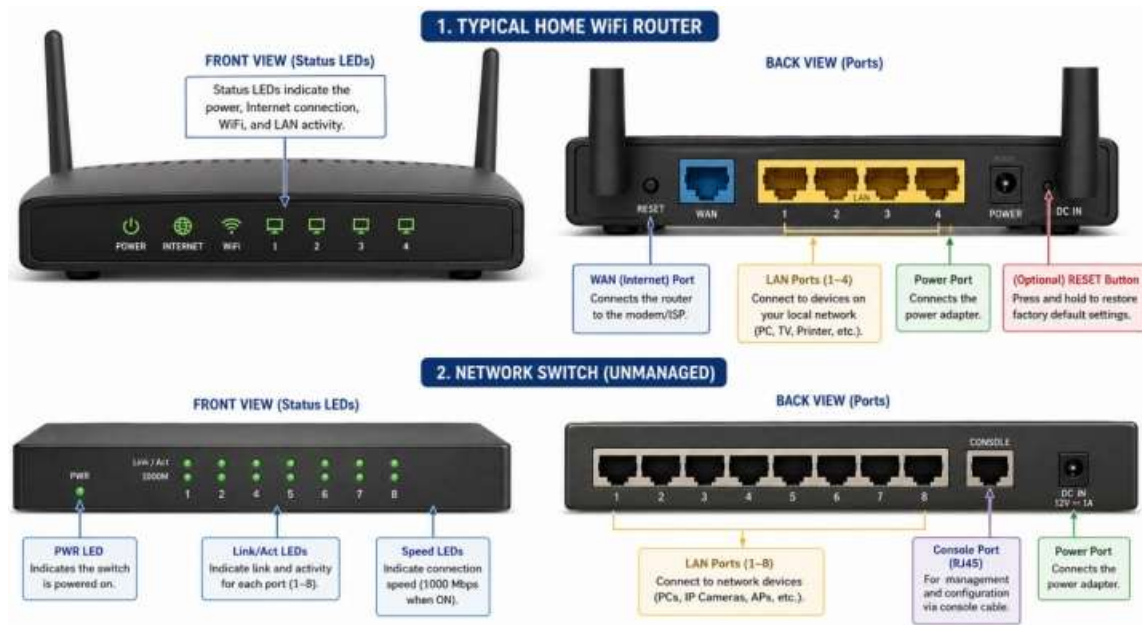


ASSIGNMENT 3.2

Q1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



A WiFi router connects the home network to the Internet through the **WAN port**. The **LAN ports** connect computers and other devices. The **status LEDs** show power, Internet, WiFi, and network activity.

A network switch connects multiple devices in the same network. It has several **LAN ports** for computers and other devices. The **Console port** is used for management, and the **status LEDs** show power, link, and activity.

Q2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

	Router	Switch
Main function	Connects separate networks, such as a home LAN to the internet, and chooses paths for packets between them.	Connects devices within the same local network, such as computers and printers in a LAN.
Address used	Uses destination IP addresses and routing tables to decide where packets should go.	Uses destination MAC addresses to forward Ethernet frames to the correct device or port.
Traffic handled	Moves traffic between different networks, such as LAN-to-WAN or between different IP subnets.	Forwards traffic within a LAN, typically sending it only to the port associated with the destination device rather than to every device.

Q3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

Types of Network Cables

1. Ethernet cable

Common use: Ethernet cables with RJ45 connectors are commonly used to connect computers, printers, routers, switches, and other devices within a wired local-area network (LAN).



Image source: [Binder USA—RJ45 Ethernet connecting cable](#)

2. Coaxial cable

Common use: Coaxial cables are commonly used for cable television, cable internet connections, satellite TV, antennas, and some CCTV systems

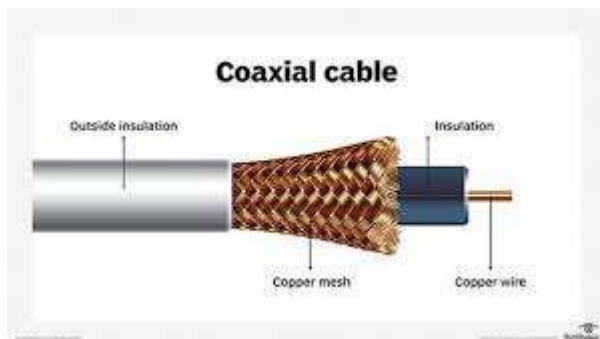


Image source: [TrueCable—RG6 compression F connectors](#)

3. Fiber-optic cable

Common use: Fiber-optic cables are commonly used for high-speed broadband, telecommunications backbones, data-center links, and long-distance internet connections.



Image source: [Rosenberger—Types of fiber-optic connectors](#)

Q4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Difference	Router	Switch
Main use	Connects different networks, such as a home LAN to the internet or a WAN.	Connects multiple devices within the same local area network (LAN).
Typical location	Usually placed at the edge of a network, between the local network and the internet/modem.	Usually placed inside a LAN, such as in an office, server room, or near connected devices.
Address used	Uses destination IP addresses and a routing table.	Uses destination MAC addresses and a MAC-address table.
How it handles data	Routes IP packets between different networks and selects an appropriate path.	Forwards Ethernet frames within the local network, normally to the specific device's port.
Network connection	Connects separate IP networks or subnets, such as LAN-to-WAN.	Connects computers, printers, servers, and other devices on the same subnet or LAN.
OSI layer	Primarily works at Layer 3, the Network layer.	Primarily works at Layer 2, the Data Link layer.